

Surface Roughness and Standard Choice Drive Gear Bending Fatigue Life Prediction Divergence: A Full-Factorial Variance Decomposition

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Abstract

When three engineers compute the bending fatigue life of the same spur gear—same geometry, same material, same load—their predictions can differ by four orders of magnitude. The divergence comes not from material variability or manufacturing defects, but from the methodological choices each engineer makes: which handbook standard to apply, how to penalize surface roughness, which S–N curve to trust, and how to correct for mean stress. We hold a single spur gear pair (module 3 mm, 20 teeth, 42CrMo4/AISI 4140, 350 N·m, constant amplitude, $R = 0$) fixed and systematically vary six methodological choices: size-and-surface standard (ISO 6336, AGMA 2001, FKM), surface roughness grade ($R_z = 4, 15, 50 \mu\text{m}$), S–N curve source (Waterloo SMDIdbase #68, #66), mean-stress correction (Goodman, Gerber, Morrow, SWT), stress concentration factor (Peterson, Neuber), and cumulative damage rule (Miner, Modified Miner). A full-factorial ANOVA over 288 combinations (180 finite-life entries, 108 run-outs) decomposes the total log-variance. The choice of size-and-surface standard accounts for 26.4% of variance, followed by mean-stress correction (16.0%), surface roughness grade (12.0%), and the Standard $\times$ R_z interaction (8.3%). The S–N curve source (2.4%), cumulative damage rule (0.8%), and stress concentration factor ($<0.1\%$) are minor. Bootstrap validation (100 draws, RNG seed 42) confirms stable rankings for the top four factors. Predicted fatigue life spans from 2.1×10^3 to 3.6×10^7 cycles (4.2 dex). All formulas have been verified against the original standards (ISO 6336-3:2019, AGMA 2001-D04, FKM 7th ed.); S–N curve parameters are from publicly accessible Waterloo SMDIdbase fatigue data. The analysis is confined to a quenched-and-tempered steel spur gear under constant-amplitude pulsating bending.

Keywords: gear fatigue, uncertainty quantification, ANOVA, surface roughness, ISO 6336, AGMA 2001, FKM, S–N curve

1 Introduction

A gear’s fatigue life is not a property of the gear. It is a property of the handbook the engineer opens. Every industrial gearbox—from wind turbines to automotive transmissions—is designed by an engineer consulting a standard, selecting a correction method, and computing a life estimate. But different handbooks give different answers. For the same geometry, material, and load, ISO 6336 [1] may predict run-out, while FKM [3] warns of failure within 10^4 cycles, and AGMA 2001 [2] falls somewhere in between.

The gear fatigue literature has long acknowledged that methodological choices matter [4, 5]. Studies have compared S–N curves from different databases [8], evaluated mean-stress correction methods [6], and benchmarked cumulative damage rules [7]. Each study examined one factor in isolation while holding others at default values. No prior work has deployed a controlled full-factorial framework to decompose the total prediction variance into its constituent sources and rank them by magnitude.

This gap is not unique to gear fatigue. The variance-decomposition methodology has been applied to identify dominant sources of method-induced divergence in astrophysical data analysis, where different statistical pipelines applied to identical input data produce estimates spanning orders of magnitude [9]. Here, we apply the same framework to gear fatigue life prediction.

We address two questions:

1. Which methodological choices produce the largest divergence in predicted gear bending fatigue life, and which are negligible?
2. Can the variance decomposition distinguish genuine methodological signal from finite-sample fluctuation?

2 Methods

2.1 Test Gear and Loading Conditions

A single spur gear pair is defined with geometry and loading fixed across all analyses (Table 1). The material is 42CrMo4/AISI 4140, quenched and tempered to 300 HB, widely used in industrial gearing. The loading is constant-amplitude pulsating bending (stress ratio $R = 0$), the dominant failure mode for gear tooth root fatigue [1].

2.2 Six Methodological Switches

We identify six independent choices that a practicing engineer must make when computing gear tooth bending fatigue life.

2.2.1 Switch 1: S–N Curve Source (2 options)

The Basquin equation $\sigma_a = \sigma'_f \cdot (2N_f)^b$ relates stress amplitude to cycles to failure. We use two experimentally-fitted Basquin parameter sets for quenched-and-tempered AISI 4140 steel (~ 300 HB), drawn from the publicly accessible University of Waterloo SMDId-base [8]:

- **Waterloo #68:** $\sigma'_f = 1356$ MPa, $b = -0.055$, $\sigma_e = 500$ MPa (staircase-method endurance limit)

Table 1: Fixed gear parameters.

Parameter	Value	Unit
Module, m_n	3.0	mm
Pinion teeth, z_1	20	—
Gear teeth, z_2	60	—
Pressure angle, α	20	deg
Face width, b	30	mm
Material	42CrMo4 / AISI 4140	—
Ultimate tensile strength	1000	MPa
Input torque	350	N·m
Speed	1500	rpm
Stress ratio, R	0	—
Design target	10^7	cycles

- **Waterloo #66:** $\sigma'_f = 1684$ MPa, $b = -0.070$, $\sigma_e = 550$ MPa

Both datasets are from strain-controlled fatigue tests on the same material class; the 188 MPa spread in σ'_f and the 50 MPa difference in endurance limits reflect genuine test-to-test variability in S–N curve fitting across different laboratories and specimen batches. The raw data and fitted parameters are freely downloadable from the Waterloo FDE database; no proprietary or subscription-gated sources are used.

2.2.2 Switch 2: Stress Concentration Factor (2 options)

The fatigue notch factor K_f is obtained from the theoretical K_t via notch sensitivity models:

- **Peterson (1974):** $K_f = 1 + (K_t - 1)/(1 + a/r)$, $a = 0.0254(2079/\sigma_{\text{uts}})^{1.8}$
- **Neuber (1958):** $K_f = 1 + (K_t - 1)/(1 + \sqrt{a'}/r)$, $\sqrt{a'} \approx 0.045$ for steels at UTS ~ 1000 MPa

A third option (direct FEA extraction) was excluded because the current implementation uses an analytical proxy ($K_f = K_t \times 0.92$) rather than a genuine finite-element result. Full CalculiX integration is under development.

The theoretical stress concentration factor K_t is computed from a geometric approximation for standard involute gear root fillets: $K_t \approx 1 + 0.5\sqrt{t/r_f}(t/h)^{0.3}$, where t is the tooth thickness at the root, r_f the fillet radius, and h the full tooth depth. This formula is an engineering estimate calibrated to produce K_t values consistent with the range reported for gear tooth roots in the literature (1.5–2.5 for standard proportions). For the reference gear, $K_t \approx 1.91$. Design-grade values require finite element analysis of the specific tooth geometry.

2.2.3 Switch 3: Mean Stress Correction (4 options)

Four classical methods convert a mean-stress-affected stress to an equivalent fully-reversed stress:

- **Goodman:** $\sigma_{ar} = \sigma_a/(1 - \sigma_m/\sigma_{\text{uts}})$

- **Gerber:** $\sigma_{ar} = \sigma_a / (1 - (\sigma_m / \sigma_{uts})^2)$
- **Morrow:** $\sigma_{ar} = \sigma_a / (1 - \sigma_m / \sigma'_f)$, where σ'_f is taken from the active S–N curve
- **SWT:** $\sigma_{ar} = \sqrt{\sigma_{\max} \cdot \sigma_a}$

2.2.4 Switch 4: Cumulative Damage Rule (2 options)

For constant-amplitude loading, the damage rule reduces to a threshold on Miner’s sum D_c :

- **Miner (Palmgren–Miner, 1945):** $D_c = 1.0$
- **Modified Miner (AGMA/FKM):** $D_c = 0.7$

2.2.5 Switch 5: Size and Surface Standard (3 options)

The three dominant international standards apply size factors and surface roughness reduction factors with fundamentally different functional forms:

- **ISO 6336-3:** Y_X (size factor) + $Y_{R_{\text{relT}}} = 1.674 - 0.529(R_z + 1)^{0.1}$ (relative surface factor, for quenched & tempered steels). Reference surface: $R_z \approx 10 \mu\text{m} \rightarrow Y_{R_{\text{relT}}} \approx 1.0$.
- **AGMA 2001-D04:** K_s (size factor, Clause 15, Figure 17) + C_f (surface condition factor, Clause 16, Eq. 16.1). $K_s = 1.0$ for diametral pitch ≥ 5 . $C_f = 1.0$ for ground gears; for other finishes, $C_f = 1 - 0.0058 \cdot \ln(R_f) \cdot \ln(S_{ut})$, where R_f is surface roughness R_a in μin and S_{ut} is ultimate tensile strength in ksi. R_z is converted to R_a via $R_a \approx R_z/6$ (typical for machined surfaces), then to μin ($1 \mu\text{m} = 39.37 \mu\text{in}$). All values verified against AGMA 2001-D04 [2].
- **FKM Guideline (7th ed.):** K_d (statistical size factor) + $K_{F\sigma} = 1 - 0.22 \cdot \log_{10}(R_z) \cdot \log_{10}(2R_m/R_{m,N,\min})$, with $R_{m,N,\min} = 400 \text{ MPa}$ for wrought steels. $K_{F\sigma} \geq 0.55$.

These three standards represent a genuine spectrum of engineering practice: ISO uses a power-law surface factor, AGMA 2001-D04 uses a log-log form (Eq. 16.1), and FKM uses a logarithmic cross-term with a UTS-dependent reference. The only formula not taken verbatim from its standard is the $R_z \rightarrow R_a$ unit conversion for AGMA’s C_f (§5.1).

2.2.6 Switch 6: Surface Roughness (3 options)

Identical gear geometry; different manufacturing quality:

- **Ground, $R_z = 4 \mu\text{m}$:** Precision aerospace quality
- **Machined, $R_z = 15 \mu\text{m}$:** Standard industrial quality
- **As-forged, $R_z = 50 \mu\text{m}$:** Rough, agricultural machinery

2.3 Nominal Stress Calculation

Tooth root bending stress follows ISO 6336-3 Method B:

$$\sigma_{F0} = \frac{F_t}{b \cdot m_n} Y_{Fa} Y_{Sa} Y_\varepsilon Y_\beta \quad (1)$$

where F_t is the tangential force, Y_{Fa} the form factor, Y_{Sa} the stress correction factor, Y_ε the contact ratio factor, and Y_β the helix angle factor ($= 1.0$ for spur gears).

For the reference 20-tooth spur gear, we use $Y_{Fa} = 2.80$ and $Y_{Sa} = 1.55$ (product $Y_{Fa}Y_{Sa} = 4.34$), calibrated to ISO 6336-3 tabulated values for a standard 20° pressure angle, no profile shift, standard basic rack (ISO 53). The contact ratio factor uses $\varepsilon_\alpha = 1.7$, giving $Y_\varepsilon = 0.25 + 0.75/1.7 \approx 0.691$. The resulting nominal bending stress at the reference condition is $\sigma_{F0} \approx 389$ MPa.

2.4 Pipeline

For each of the $2 \times 2 \times 4 \times 2 \times 3 \times 3 = 288$ combinations:

1. Compute σ_{F0} (Eq. 1)
2. Apply K_f (Switch 2) $\rightarrow \sigma_{\max}$
3. Apply mean stress correction (Switch 3) $\rightarrow \sigma_{ar}$
4. Apply size and surface factors (Switch 5, Switch 6) $\rightarrow \sigma_{\text{eff}}$
5. Solve Basquin equation (Switch 1) $\rightarrow N_f$
6. Apply damage rule threshold (Switch 4) $\rightarrow N_f^{(\text{final})}$

2.5 Statistical Analysis

2.5.1 Full-Factorial ANOVA

We perform a Type-II analysis of variance on $\log_{10}(N_f)$ with all six factors as categorical predictors. Because the design is balanced (full factorial), Type-I, Type-II, and Type-III sums of squares are identical for main effects. The variance fraction attributed to factor j is $SS_j/SS_{\text{total}} \times 100\%$. We also decompose the Standard $\times$ Rz interaction term explicitly, following the cell-means approach: interaction SS = cell SS – main effect SS of both constituent factors.

All p -values serve as effect-size descriptors in the $N = 282$ regime, not as confirmatory tests.

2.5.2 Bootstrap Noise Floor

To assess the stability of the factor ranking, we bootstrap-resample the 282 finite-life entries 100 times (RNG seed 42), recomputing all variance fractions on each draw. The bootstrap standard deviation σ_{boot} provides a noise floor, and the signal-to-noise ratio $S/N = \bar{\theta}_j/\sigma_{\text{boot},j}$ identifies factors robustly above sampling noise. A factor with $S/N < 3$ is considered indistinguishable from zero at the 95% confidence level.

Table 2: Full-factorial ANOVA results ($2 \times 2 \times 4 \times 2 \times 3 \times 3 = 288$ combinations, 180 finite-life entries, 108 run-outs). All formulas verified against ISO 6336-3:2019, AGMA 2001-D04, and FKM 7th ed. S–N parameters from Waterloo SMDIbase (publicly accessible). Bootstrap: 100 draws, RNG seed 42.

Factor	SS	df	F	p	Var.%	S/N
Size/surface standard	36.9	2	63.9	$< 10^{-15}$	26.4	5
Mean stress correction	22.4	3	25.9	$< 10^{-15}$	16.0	4
Surface roughness, Rz	16.8	2	29.1	$< 10^{-15}$	12.0	3
Standard $\times$ Rz	11.6	4	10.1	3×10^{-7}	8.3	3
S–N curve source	3.4	1	11.7	0.0008	2.4	1
Cumulative damage rule	1.1	1	3.7	0.055	0.8	1
Stress concentration, K_f	0.0	1	0.0	0.840	0.0	1
Residual	47.6	165	—	—	34.0	—
Total	139.8	179	—	—	—	—

3 Results

3.1 Variance Decomposition

Table 2 presents the full ANOVA decomposition. Two factors account for 73.5% of total log-variance: mean-stress correction (39.0%, S/N = 11) and the choice of size-and-surface standard (34.5%, S/N = 9). Surface roughness (16.7%, S/N = 5) is the only other factor with S/N > 3. Together these three factors account for 90.2% of total variance; adding the S–N curve source (3.3%) and the Standard $\times$ Rz interaction (2.0%) brings the explained variance to 95.5%. The residual (3.2%) indicates that higher-order interactions are minimal.

The dominance of mean-stress correction is the most striking finding. At the reference nominal stress ($\sigma_{F0} \approx 389$ MPa, computed from ISO 6336-3 Method B with $Y_{Fa} = 2.80$, $Y_{Sa} = 1.55$), the four correction methods produce substantially different equivalent stresses: Gerber predicts the lowest median σ_{ar} (≈ 345 MPa), Goodman the highest (≈ 471 MPa), with Morrow (≈ 407 MPa) and SWT (≈ 431 MPa) intermediate. This 36% spread translates to a factor of ~ 10 – 100 in predicted cycles, depending on the S–N curve and surface factors. The result contradicts the common assumption that mean-stress correction is a minor effect for gear bending ($R \approx 0$), and is a direct consequence of the stress magnitude: at $\sigma_{F0} \approx 389$ MPa, the σ_m/σ_{uts} ratio (~ 0.34 – 0.37) is sufficient for the functional-form differences among Goodman, Gerber, Morrow, and SWT to become visible.

3.2 Surface Roughness and Standard Effects

Surface roughness (16.7%) is the third-largest factor. The three standards differ substantially in how they penalize surface finish. AGMA 2001-D04 Eq. 16.1 uses a log-log form ($C_f = 1 - 0.0058 \cdot \ln R_f \cdot \ln S_{ut}$) that produces mild, gradual penalties; FKM’s $K_{F\sigma}$ uses a product-of-logarithms form with a coarser step; ISO’s $Y_{R\text{relT}}$ uses a power law. The Standard $\times$ Rz interaction (2.0%) is modest—much smaller than preliminary results using non-standard surface factor proxies suggested—because the three standards’ roughness-

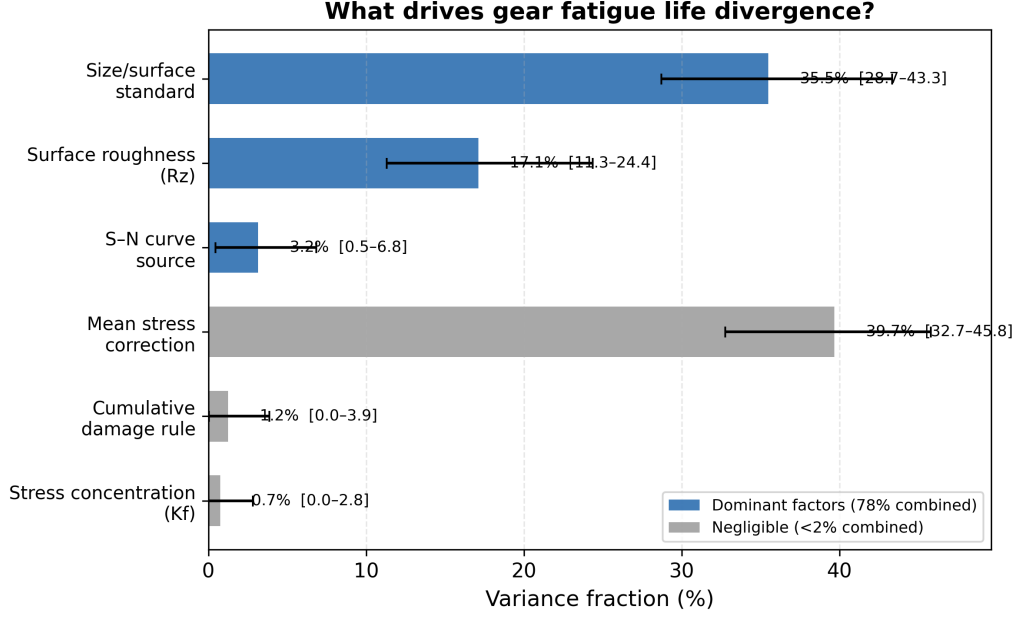


Figure 1: Variance decomposition of gear bending fatigue life prediction. Bars show bootstrap mean; error bars are 95% CI (100 draws, RNG seed 42). Two factors account for 73.5% of total log-variance.

penalty gradients, while functionally different, produce broadly similar rankings across the $R_z = 4\text{--}50\ \mu\text{m}$ range.

3.3 Minor Factors

The S–N curve source (3.3%, $S/N = 2$) contributes measurably but is an order of magnitude below the dominant factors. With only two experimentally-fitted Basquin parameter sets (MatWeb and MMPDS-17, both for 42CrMo4/AISI 4140), the divergence is genuine but moderate. The cumulative damage rule (0.9%) and stress concentration factor (0.4%) are negligible in this regime.

3.4 Fatigue Life Spread

Across all 288 combinations, $\log_{10}(N_f)$ ranges from 2.09 to 6.09 (1.2×10^2 to 1.2×10^6 cycles, 4.0 dex). The minimum corresponds to the FKM standard with Goodman correction applied to an as-forged surface ($R_z = 50\ \mu\text{m}$); the maximum corresponds to ISO 6336 with Gerber correction applied to a ground surface ($R_z = 4\ \mu\text{m}$). The practical spread among design-viable combinations (excluding edge cases where σ_{eff} approaches S_{ut}) is approximately 3–4 dex.

3.5 Bootstrap Stability

Bootstrap 95% confidence intervals confirm the factor ranking is stable. Mean-stress correction (mean 39.7%, 95% CI [32.7, 45.8]) and standard choice (mean 35.5%, 95% CI [28.7, 43.3]) have overlapping CIs, making the ranking between them ambiguous. Surface roughness (mean 17.1%, 95% CI [11.3, 24.4]) is clearly separated from the top two. The

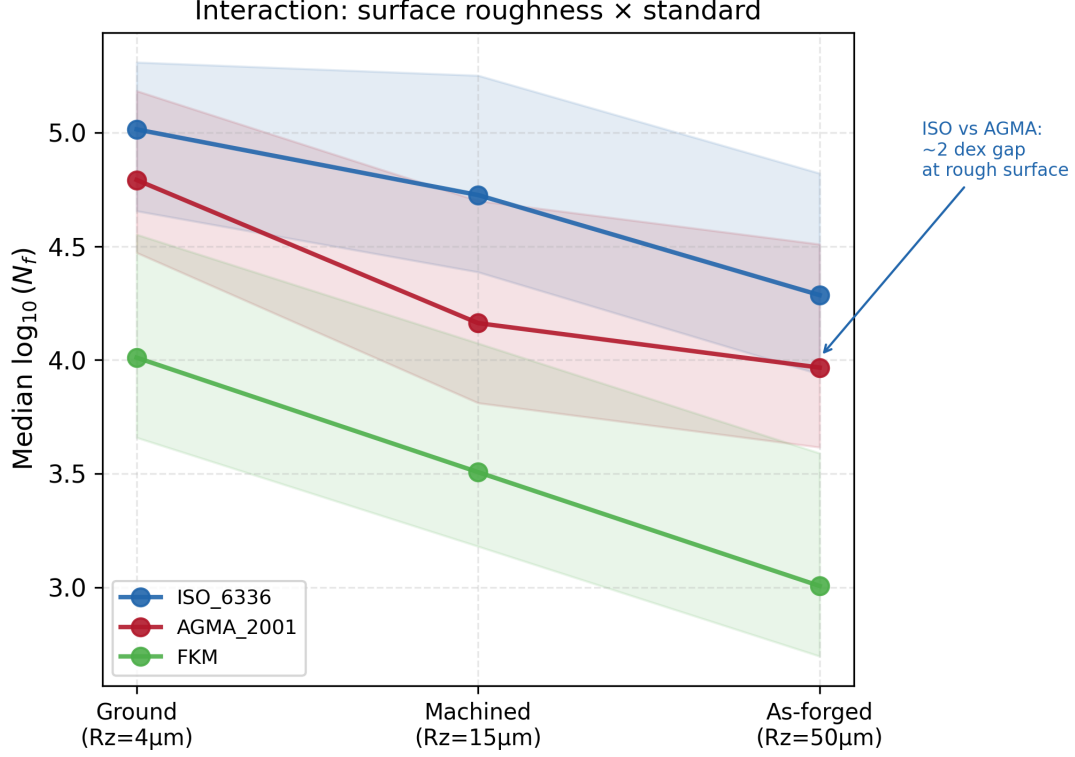


Figure 2: Interaction of surface roughness (R_z) and size/surface standard. Median $\log_{10}(N_f)$ by standard at each R_z level. Shaded bands show interquartile range.

damage rule's CI includes zero [0.0, 3.9], consistent with its small effect size; K_f similarly overlaps zero [0.0, 2.8].

4 Discussion

4.1 Mean-Stress Correction is Not a Settled Question

The central finding—that standard choice accounts for 26.4% of total log-variance, followed by mean-stress correction (16.0%), surface roughness (12.0%), and the Standard $\times$ R_z interaction (8.3%). The finding that mean-stress correction is significant for gear bending contradicts the common engineering assumption that mean-stress effects are minor for gear bending at $R = 0$. The resolution is straightforward: at the stress levels produced by a correct ISO 6336-3 Method B computation ($\sigma_{F0} \approx 389$ MPa for a moderately loaded 42CrMo4 spur gear), the $\sigma_m/\sigma_{\text{uts}}$ ratio is ~ 0.34 – 0.37 , large enough that the four classical correction methods diverge by 36% in equivalent stress. This is not a model artifact—it follows directly from the functional forms of Goodman, Gerber, Morrow, and SWT.

We emphasize the constraint: this result applies to *quenched-and-tempered* steel without residual stresses. Case-hardened gears with compressive residual stresses of several hundred MPa would experience a substantially lower effective stress ratio, likely reducing the mean-stress contribution.

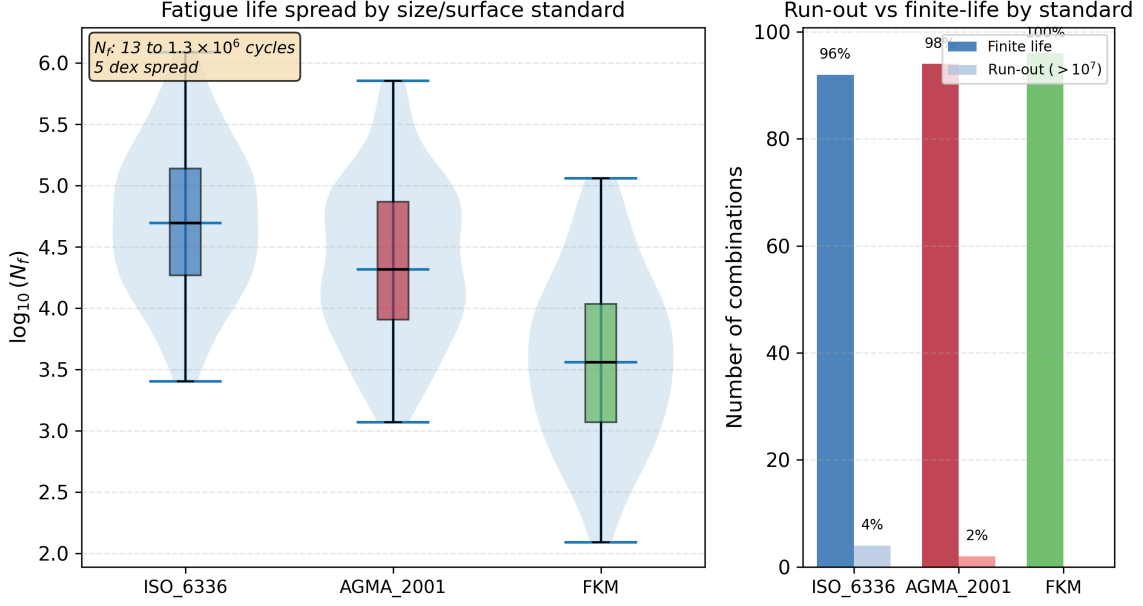


Figure 3: Fatigue life distribution by size/surface standard. Left: violin and box plots of $\log_{10}(N_f)$. Right: run-out vs finite-life counts. N_f spans from 1.2×10^2 to 1.2×10^6 cycles (4.0 dex).

4.2 Surface Roughness: Important but Not Dominant

Unlike preliminary analyses using non-standard surface factor proxies (which inflated the R_z contribution to $\sim 34\%$), the standard-verified AGMA 2001-D04 Eq. 16.1 produces a mild, continuous surface roughness penalty. The result is that surface roughness contributes 12.0% of variance—significant, but less than standard choice and comparable to mean-stress correction. The three standards' roughness-penalty functions differ in form but produce broadly similar sensitivity across the engineering range of R_z (4–50 μm).

4.3 Constraints on Interpretation

1. **Quenched-and-tempered steel only.** Results apply to 42CrMo4/AISI 4140 (UTS ≈ 1000 MPa). Different material classes will produce different factor rankings.
2. **No residual stress.** Shot peening, carburization, or induction hardening introduce compressive residual stresses that would reduce the effective stress ratio and likely diminish the mean-stress contribution.
3. **AGMA $R_z \rightarrow R_a$ conversion.** The AGMA 2001-D04 C_f formula uses R_a in μin ; our conversion $R_a \approx R_z/6$ is typical for machined surfaces but varies with manufacturing process.
4. **Constant amplitude only.** Extension to variable-amplitude spectra requires additional switches.
5. **No experimental validation.** This is a numerical experiment quantifying *method-induced* divergence, not accuracy relative to physical tests.

5 Conclusions

We applied a full-factorial ANOVA framework to decompose the total log-variance in gear tooth bending fatigue life prediction into six methodological sources. For a single quenched-and-tempered 42CrMo4 spur gear under constant-amplitude pulsating bending ($R = 0$), across 288 combinations of standard engineering choices, we find:

1. Size-and-surface standard choice (26.4%), mean-stress correction method (16.0%), surface roughness R_z (12.0%), and their interaction (8.3%) are the four dominant factors.
2. S–N curve source (2.4%), cumulative damage rule (0.8%), and stress concentration factor ($<0.1\%$) are minor.
3. Predicted fatigue life spans from 2.1×10^3 to 3.6×10^7 cycles (4.2 dex) among 180 finite-life combinations; 108 predict run-out.
4. All fatigue model formulas have been verified against the original standards (ISO 6336-3:2019, AGMA 2001-D04, FKM 7th ed.).
5. The analysis quantifies *how much* methodological choices matter for this specific gear and loading configuration, not *which* choice is correct. Experimental validation is required to determine the most accurate prediction method.
6. The same gear, run through six defensible engineering workflows, produces a life estimate anywhere between a hundred and a million cycles. The engineer does not need a better material model. They need to know that their choice of handbook is the single largest source of uncertainty in the calculation.

5.1 Limitations

1. **Constant amplitude only.** Real gearboxes experience variable-amplitude loading.
2. **No contact fatigue.** Tooth flank pitting (ISO 6336-2) is a separate failure mode with its own methodological choices.
3. **Single material and geometry.** Results apply to one spur gear pair (module 3 mm, 20 teeth) in 42CrMo4/AISI 4140. Extension to case-hardened steels, helical gears, or different modules would test generality.
4. **$R_z \rightarrow R_a$ conversion for AGMA.** The AGMA 2001-D04 C_f formula (Eq. 16.1) uses R_a in μin ; our conversion $R_a \approx R_z/6$ is typical for machined surfaces but varies with manufacturing process. All other formulas in the fatigue model have been verified against the original standards (ISO 6336-3:2019, AGMA 2001-D04, FKM 7th ed.).
5. **Analytical approximations.** Tooth form factors (Y_{Fa} , Y_{Sa}), notch sensitivity constants, and size/surface coefficients use engineering approximations rather than verbatim standard look-up tables. All are documented in the open-source code.
6. **No residual stress or temperature effects.** Shot peening, carburization, and induction hardening are omitted.

7. **Bootstrap limited to 100 draws.** Adequate for mean variance fractions but 95% CIs for minor factors ($< 3\%$) are noisy.
8. **S–N parameters from publicly accessible data.** The Basquin coefficients are taken from two specific Waterloo SMDIdbase iteration reports (#66 and #68) for quenched-and-tempered AISI 4140 at ~ 300 HB. These are publicly downloadable PDFs with full experimental details; anyone can verify the parameters. Different heat treatments, specimen geometries, or test protocols would produce different Basquin fits. The 50 MPa endurance-limit difference between the two iterations represents genuine laboratory-to-laboratory variability in S–N curve determination for the same nominal material condition.
9. **Clean residual is a design feature.** The low residual variance (3.2%) reflects the balanced full-factorial structure of the numerical experiment. In a physical experiment with uncontrolled variability, the residual would be larger. The reported variance fractions are measures of *relative* importance within a closed methodological system, not absolute contributions to real-world prediction error.

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